Class Hydrozoa

Hydrozoans

Occurrence probably 3,800 spp. worldwide; in plankton, and on sea, river, and lake beds

This group of cnidarians contains a diverse array of animals, with various body shapes and lifestyles, and sizes ranging from as small as 1⁄32 in (1 mm) to as large as 3¼ ft (1 m) across. In the course of their lives, they usually pass through both an attached polyp stage and a free-living medusa stage. Most are colonial, containing tens to thousands of polyps. Some colonies have hard skeletons and resemble corals, while others are often mistaken for jellyfish, drifting through the surface waters. Colonial forms that remain on the seabed mostly have polymorphic polyps, with different polyps being shaped for different tasks. The result is a colony that looks and works like a single animal. Planktonic colonies may also have polymorphic polyps, with some catching prey and others digesting the catch. Mostly carnivorous, hydrozoans use their tentacles to capture organisms ranging from minute planktonic animals to sizable fishes; some are dependent on symbiotic photosynthetic algae. Most animals in this class—which includes the sea fir and by-the-wind-sailor, as well as the freshwater hydras—are harmless to humans, but a few, including the notorious Portuguese man-of-war, have potentially deadly stings. They are preyed on by some mollusks, gastropods, polychaetes, sea spiders, starfish, and fishes. Reproduction occurs both sexually and asexually.

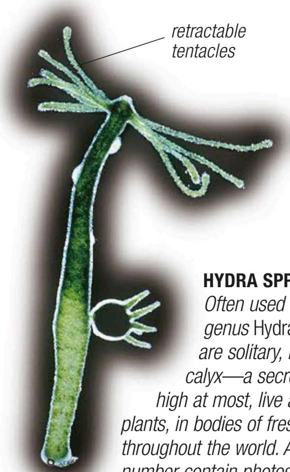
PORTUGUESE MAN-OF-WAR
Found almost worldwide in tropical and temperate seas, the Portuguese man-of-war, Physalia physalis *(right), has a well-deserved reputation for being dangerous. It looks like a single animal, but it is actually a highly integrated colony of polyps, some of which pack a powerful sting. A large, gas-filled float keeps the colony at the surface, and supports trailing tentacles up to 66 ft (20 m) long. These tentacles contain polyps that catch prey, ones that digest food, and others that enable the colony to reproduce.*

BY-THE-WIND-SAILOR
The coin-shaped Velella velella*—4 in (10 cm) in diameter—is a highly modified hydroid polyp that, like the Portuguese man-of-war, drifts on the ocean surface. It is moved by the wind acting on the small sail attached to the float and, after storms, is often washed up on tropical shores.*

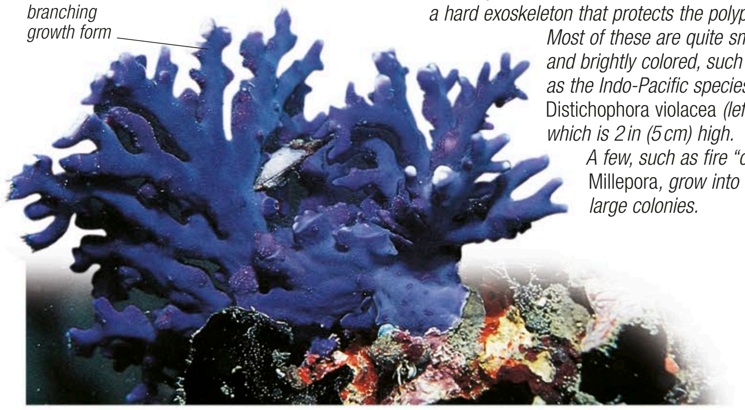
POLYORCHIS SP.
This delicate hydrozoan medusa has a transparent bell up to 1½ in (4 cm) across, with about 90 tentacles trailing from it. Common on the west coast of North America, it feeds on planktonic animals.

FEATHER HYDROIDS
Species in the widespread genus Aglaophenia *live in shallow water, and their dead remains are often washed up by the tide. Each colony can be up to 23½ in (60 cm) high, and contains numerous feeding and stinging polyps, as well as larger reproductive polyps—the solid yellow structures in the picture (right)—which release floating larvae into the water.*

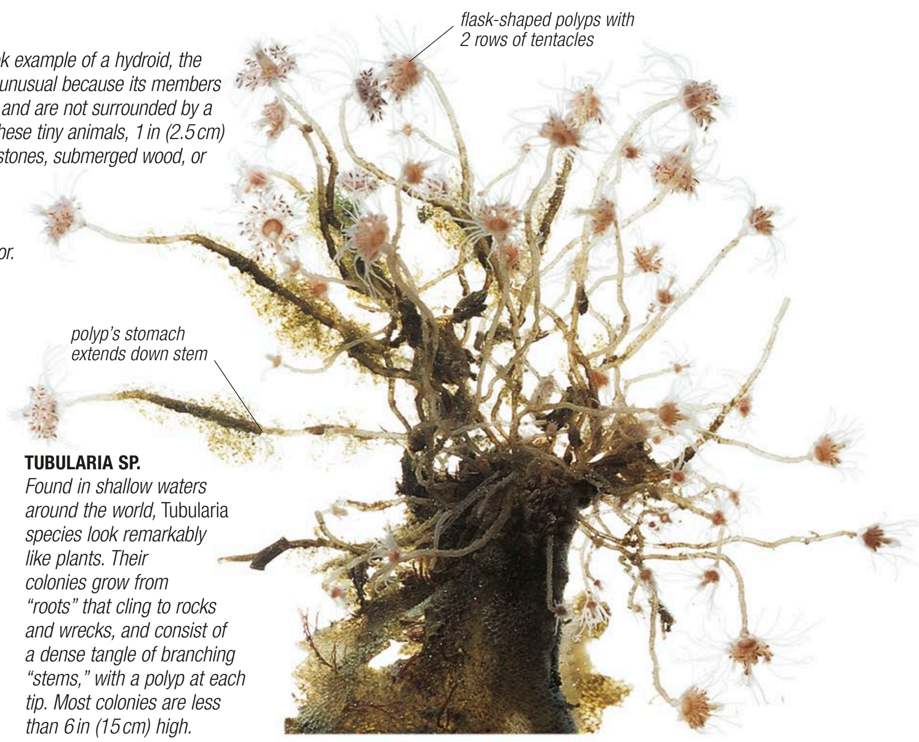
HYDRACTINIA SPP.
Members of the genus Hydractinia*, distributed worldwide, are small, colonial hydroids, typically ¾ in (2 cm) high, that often form encrustations on animals. The hydroids get improved access to food by being carried around, and in return, protect their host with their stings.* Hydractinia echinata *grows on shells of hermit crabs (see above).*



HYDRA SPP.
Often used as a textbook example of a hydroid, the genus Hydra *is, in fact, unusual because its members are solitary, not colonial, and are not surrounded by a calyx—a secreted tube. These tiny animals, 1 in (2.5 cm) high at most, live attached to stones, submerged wood, or plants, in bodies of freshwater throughout the world. A large number contain photosynthetic algae, which give them a green color.*



DISTICHOPORA VIOLACEA
A few hydrozoans resemble corals and have a hard exoskeleton that protects the polyps. Most of these are quite small and brightly colored, such as the Indo-Pacific species Distichophora violacea *(left), which is 2 in (5 cm) high. A few, such as fire "coral"* Millepora*, grow into very large colonies.*



TUBULARIA SP.
Found in shallow waters around the world, Tubularia *species look remarkably like plants. Their colonies grow from "roots" that cling to rocks and wrecks, and consist of a dense tangle of branching "stems," with a polyp at each tip. Most colonies are less than 6 in (15 cm) high.*